Ex-5

[Verification of Norton's Theorem]

13/2/26

Aim ÷ To verify Norton's theorem practically and theoretically for the given DC circuit.

Apparatus required ÷

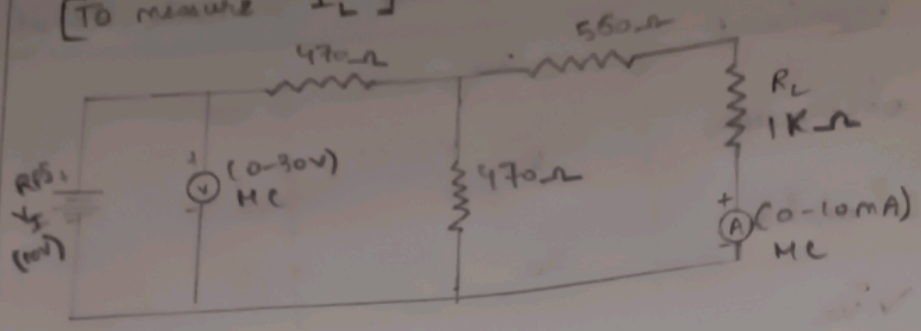
S.NO	Apparatus	Specification	Qty
1.	Regulated Power	(0-30 v)	1
2.	voltmeter	(0-30V)	1
3.	Ammeter	(0-10mA)	1
4.	Resistor	470Ω, 560Ω, 1KΩ	2, 1, 1
5.	Breadboard	–	1
6.	Multimeter	–	1

Theory ÷

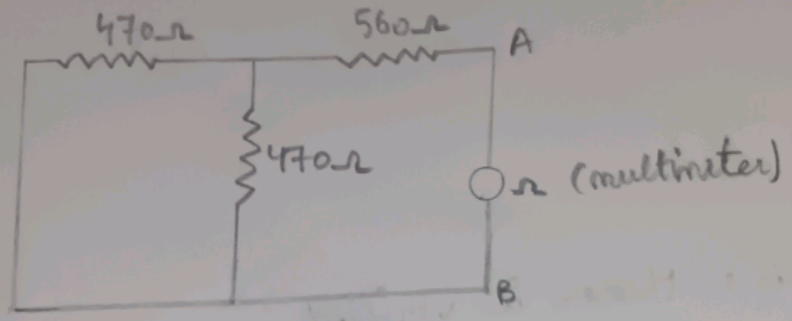
Norton's theorem : A linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source, in parallel with a resistor $R_p (= R_{TH})$ where I_N is the short circuit current through the load terminal and R_N is the equivalent resistance at the load terminals.

Norton's resistance (R_N) is the look resistance across the load terminal when all the sources are replaced by their internal resistance. An ideal voltage source is replaced by short circuiting as is internal resistance is zero.

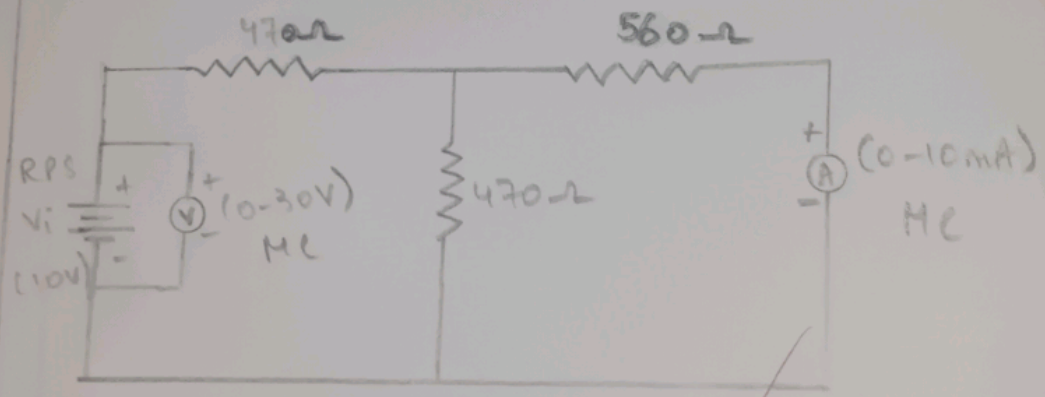
[To measure I_L]



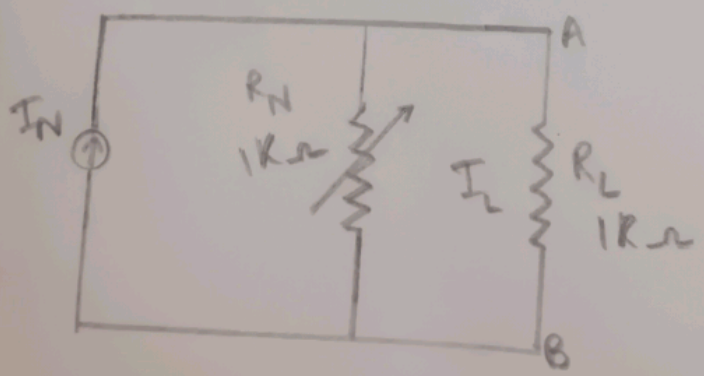
[To measure R_{TH} or R_N]



[To measure I_N or I_{sc}]



[Norton's equivalent circuit]



[T. column 1]

For I_L :

V_1 (volts)	I_L
10 V	275 mA

[T. column 2]

For R_N :

V_1 (volts)	R_{TH} (Ω)
10 V	77

[T column 3]

For I_N or (I_{sc}) :

V_1 (volts)	I_N (A)
10 V	6.27

Calculation :

Parallel :

R_{eq} 1 = 255.5 Ω

R_{eq} 2 = 235 Ω

R_{Total} = 255.5 + 470 Ω = [725.5 Ω]

$I_s = V/R_I = 10/25.5$ = [0.013 A]

Procedure :

1.) Make the connections as per circuit diagram.

2.) Vary the RPS and set an input of 10V.

3.) Note down the Voltmeter reading and ammeter reading in tabular column.

4.) Switch off the supply and make connections for the circuit diagram 2.

5.) Measure the thevinin's resistance (R_{TH}) = Norton's resistance (R_N).

6.) Set an input voltage of 10V on RPS and note down the voltage meter reading (V and V_{TH}).

7.) Switch off the supply and make connections for circuit diagram.

8.) Set an input voltage of 10V in the RPS and take the voltmeter reading V_i and ammeter reading $I_N (\equiv I_{sc})$.

9.) Calculate the i_l value using the formula :

$$I_2 = V_{TH} / (R_{TH} + R_L)$$

Norton's theorem :

$$I_L = I_N * R_N / (R_N + R_L)$$

10.) Theoretically verify Norton's theorem.

$$I_N = 0.013 \times \frac{470}{470+560}$$

$$I_N = 6.76\,mA$$

To measure R_N :

$$R_{eq} = 560 + 235$$

$$R_N = R_2 + \frac{R_1 R_3}{R_1 + R_3}$$

$$= 560 + 235$$

$$= 795\,\Omega$$

Load Current (I_L) :

$$I_L = I_N \times \frac{R_N}{R_N + R_L}$$

$$= 6.75 \times \frac{795}{795+1000}$$

$$= 2.768\,mA.$$

$$V_L = I_L \times R_L$$

$$[V_L = 2.768\,V]$$

S.No	Description	Marks alloted	Marks obtain
1.	Experiment	20	20
2.	observation	20	18
3.	calculation	20	18
4.	Viva	20	18
5.	Record	20	20
6.	Total	100	94

Result :

Thus, Norton's theorem is verified.